INCH-POUND
MIL-DTL-23053/7C
17 June 1996
SUPERSEDING
MIL-I-23053/7B
01 May 1986

DETAIL SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, POLYETHYLENE TEREPHTHALATE, NON-CROSSLINKED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the sleeving described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-DTL-23053

REQUIREMENTS:

Polymer type: The base polymer used in formulating this sleeving shall be polyethylene terephthalate.

Continuous operating temperature range: -60°C (-76°F) to +135°C (+275°F).

Classification: The heat shrinkable sleeving shall be following classes, as specified (see 6.2a):

Class 1 - Single longitudinal seam, thermally bonded
Class 2 - Spirally wound, resin bonded

Color: The heat shrinkable sleeving shall be furnished in clear, as manufactured. The acquiring activity may specify colors for Class 2 sleeving only (see 6.2b). Colors shall conform to Class I of MIL-STD-104.

Longitudinal change:

Class 1: 30 percent maximum
Class 2: 35 percent maximum

Military part number: The military part number shall consist of the basic number of this specification sheet and dash numbers as follows:

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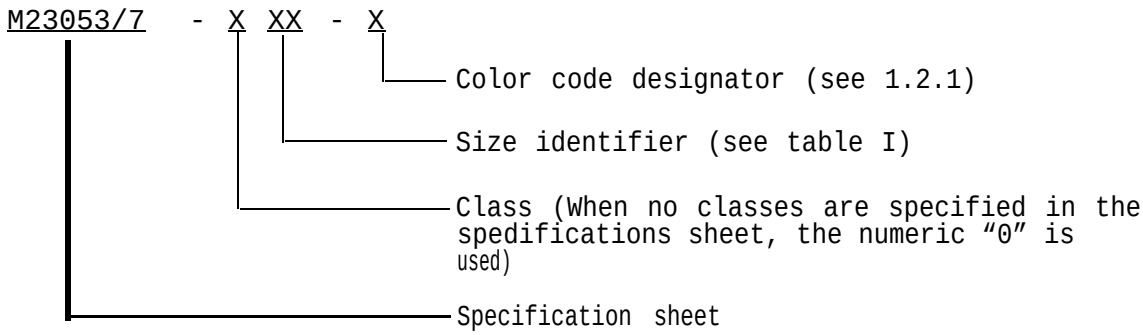

Example: Class 1, .125 (3.2 mm), as supplied ID sleeving shall be identified as M23053/7-103-C.

TABLE I. Construction details, inches (mm). 1/

Military part number 1/	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
Class 1			
M23053/7-101-C		No longer	Manufactured
M23053/7-102-C		No longer	Manufactured
M23053/7-103-C	.125(3.2)	.088(2.2)	.0027 ± .0004(.0687 ± .010)
M23053/7-104-C	.187(4.8)	.131(3.3)	.0027 ± .0004(.0687 ± .010)
M23053/7-105-C	.250(6.4)	.175(4.6)	.0027 ± .0004(.0687 ± .010)
M23053/7-106-C	.375(9.5)	.263(6.7)	.0027 ± .0004(.0687 ± .010)
M23053/7-107-C	.500(12.7)	.350(8.9)	.0027 ± .0004(.0687 ± .010)
M23053/7-108-C	.750(19.1)	.525(13.4)	.0027 ± .0004(.0687 ± .010)
M23053/7-109-C	1.000(25.4)	.700(17.8)	.0027 ± .0004(.0687 ± .010)
M23053/7-110-C	1.375(34.9)	.963(24.5)	.0027 ± .0004(.0687 ± .010)
M23053/7-111-C	1.500(38.1)	1.050(26.7)	.0027 ± .0004(.0687 ± .010)
M23053/7-112-C	2.000(50.8)	1.400(35.6)	.0027 ± .0004(.0687 ± .010)
M23053/7-113-C	2.500(63.5)	1.750(44.5)	.0027 ± .0004(.0687 ± .010)
M23053/7-114-C	3.000(76.2)	2.100(53.3)	.0027 ± .0004(.0687 ± .010)
M23053/7-115-C	4.000(101.6)	2.800(71.1)	.0027 ± .0004(.0687 ± .010)
Class 2			
M23053/7-201-*	.015(.38)	.011(.28)	3/
M23053/7-202-*	.020(.50)	.015(.38)	
M23053/7-203-*	.025(.63)	.019(.48)	
M23053/7-204-*	.030(.76)	.023(.58)	
M23053/7-205-*	.042(1.06)	.032(.81)	
M23053/7-206-*	.046(1.17)	.035(.89)	
M23053/7-207-*	.062(1.57)	.047(1.19)	
M23053/7-208-*	.093(2.36)	.070(1.78)	
M23053/7-209-*	.125(3.18)	.094(2.4)	

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TABLE I. Construction details, inches (mm). 1/ - Continued

Military	As supplied	After unrestricted shrinkage	
part number 1/	ID minimum	ID maximum	Wall Thickness 2/
M23053/7-210-*	.187(4.75)	.140(3.6)	3/
M23053/7-211-*	.250(6.35)	.188(4.8)	
M23053/7-212-*	.312(7.92)	.234(5.9)	
M23053/7-213-*	.375(9.53)	.281(8.1)	
M23053/7-214-*	.500(12.7)	.375(9.5)	
M23053/7-215-*	.625(15.88)	.469(11.9)	
M23053/7-216-*	.750(19.1)	.563(14.3)	
M23053/7-217-*	1.000(25.4)	.750(19.1)	
M23053/7-218-*	1.125(28.58)	.844(21.4)	
M23053/7-219-*	1.250(31.75)	.938(23.8)	
M23053/7-220-*	1.375(34.93)	1.032(26.2)	
M23053/7-221-*	1.500(38.1)	1.125(28.6)	
M23053/7-222-*	1.750(44.45)	1.313(33.4)	
M23053/7-223-*	2.000(50.8)	1.500(38.1)	

1/ Diameter limits for the object to be enclosed shall be no less than 85 percent of the as supplied diameter.

2/ Class 1 wall thickness measurements shall be made away from the seam.

3/ Because of irregular shrinkdown of Class 2 sleevings, wall thickness shall be determined “as supplied” as follows:

Nominal ID (as supplied]		Wall thickness range (as supplied)	
inch	(mm)	inch	(mm)
.015 to .061	(.381 to 1.55)	.0015 to .004	(.038 to .102)
.062 to .074	(1.57 to 1.88)	.0015 to .005	(.038 to .127)
.075 to .094	(1.91 to 2.39)	.0015 to .006	(.038 to .152)
.095 to .124	(2.41 to 3.15)	.0015 to .008	(.038 to .203)
.125 to 2.000	(3.18 to 50.8)	.0015 to .010	(.033 to .254)

The acquisition activity (see 6.2) may specify the exact wall thickness required within each range.

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Wall thickness tolerance
(as supplied)

Wall		Tolerance	
inch	(mm)	inch	(mm)
.0015	(.038)	* .0005	(.0127)
.0016 to .003	(.041 to .076)	* .0008	(.0203)
.0031 to .005	(.079 to .127)	* .0010	(.0254)
.0051 to .010	(.130 to .254)	* .0015	(.0381)

4/ The asterisk in the part number shall be replaced by color code designations (see 1.2.2).

Unrestricted shrinkage: Test method 4.6.2.2; 160° ± 2°C (320° ± 4°F) for 5 minutes.

TABLE II. Physical properties.

Characteristic	Requirement	Test procedure and conditions
As supplied:		
ID, minimum	Table I	4.6.3
Restricted shrinkage	No cracks	4.6.6.1.1 160° ± 2°C (320° ± 4°F)
Voltage withstand	Pass	4.6.6.2
Color stability, Class 2	Pass	4.6.15 150° ± 2°C (302° ± 4°F), 48 hours
Clarity stability, clear sleeving	Pass	4.6.16 150° ± 2°C (302° ± 4°F), 24 hours
Heat shock	No cracks, flowing or dripping	4.6.8 180° ± 2°C (356° ± 4°F)
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3
Wall thickness	Table I	4.6.3

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TABLE II. Physical properties. - Continued

Characteristic	Requirement	Test procedure and conditions
Seam strength, Class 1, PPI (Kg/m), minimum	25 (450)	4.6.2 ASTM D882, Method A
Tensile strength, psi (MPa), minimum	20,000 (138)	4.6.13 ASTM D638, 20 inches/minute
Ultimate elongation, percent, minimum	50	4.6.13 ASTM D638, 20 inches/minute
Dielectric strength, volts/roil (Kv/mm) minimum	2,500 (98.4)	4.6.2 ASTM D2671
Volume resistivity, Ohm-cm, minimum	10^{14}	4.6.2 ASTM D876
Specific gravity, minimum	1.3	4.6.2 ASTM D792
Corrosion	No corrosion	4.6.10.1 and 4.5.10.2 155 ± 2°C (311° ± 4°F), 16 hours
Low temperature flexibility	No cracking	4.6.7.1 -60° ± 1°C (-76° ± 2°F)
Water absorption, percent, maximum	0.8	4.6.2 ASTM D570, 24 hrs at 23°C
Heat resistance, properties after:		4.6.8 150° ± 2°C (302° ± 4°F), 336 hours
Tensile strength, psi (MPa), minimum	Class 1 - 18,000 (124) Class 2 - 13,000 (89.7)	4.6.13 ASTM D638, 20 inches/minute
Ultimate elongation, percent, minumum	Class 1 - 55 Class 2 - 40	4.6.13 ASTM D638, 20 inches/minute

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Table II. Physical properties. - Continued

Characteristic	Requirement	Test procedure and condition
Dielectric strength volt/roil (Kv/mm), minimum	2,500 (63.5)	4.6.2 ASTM D876
Fluid resistance, properties after:		4.6.11
Tensile strength psi (MPa), minimum	20,000 (98.4)	
Dielectric strength, volt/roil (Kv/mm), minimum	2,500(63.5)	
Flammability	Self-extin-guishing 1 minute, 25 percent maximum indicator flag burned or charred	4.6.14 Procedure B ASTM D2671
Fungus resistance	No growth	4.6.2 ASTM G21

1/ Unless otherwise specified, the requirements, test procedures and conditions are for all classes.

Storage life: Supplier shall certify to storage at 18° to 35°C (65° to 95°F) for 4 years. Conformance to 3.5. See 3.5.2 for storage life extension.

Intended use: To provide a tight protective covering for relays, electronic coils and components and small capacitor cans.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Army - EA
Navy - EC
Air Force - 99
DLA - GS

Preparing activity:
Navy - AS

(Project 5970-1129-07)

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